OpenGEMS

A lightweight open-source tool for isochronal mapping and velocity field visualization on 2D electrode grids. (Version:0.1)

Current scope

- Load a 2D activation-time matrix from CSV
- Build an interpolated isochronal map
- Estimate an apparent velocity field from the activation-time gradient
- Export figures as JPG and SVG

Input Configuration

Number of Rows (int)

10

Number of Columns (int)

10

Row Spacing (mm)

5.000

Column Spacing (mm)

5.000

Color Scale

rainbow

Activation Time Matrix (.csv)

opengem..._10x10.csv
0.7KB

Run Analysis

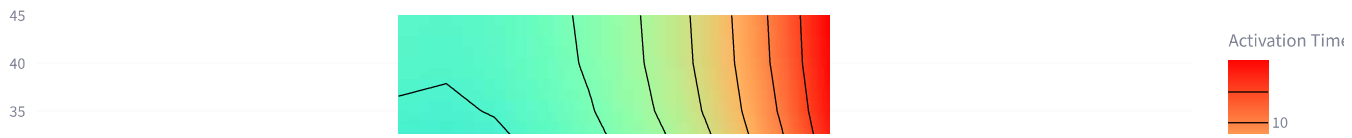
Loaded Activation Matrix

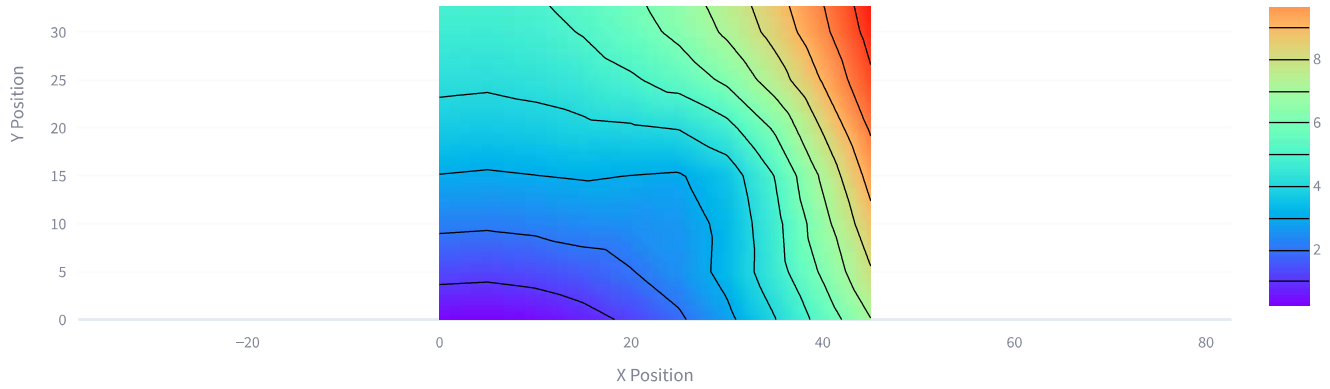
	0	1	2	3	4	5	6	7	8	9
0	0.251	0.2	0.3443	0.6733	1.1756	1.8635	2.783	3.9635	5.3756	6.9
1	1.2806	1.2264	1.3525	1.6152	1.9659	2.4567	3.2795	4.5567	6.1659	7.9
2	2.1899	2.13	2.2237	2.3679	2.4483	2.5876	3.2379	4.6876	6.6483	8.6
3	2.9795	2.9163	2.9915	3.0675	2.9926	2.9299	3.4811	5.0299	7.1926	9.3
4	3.6499	3.59	3.6837	3.8279	3.9083	4.0476	4.6979	6.1476	8.1083	10.1
5	4.2006	4.1464	4.2725	4.5352	4.8859	5.3767	6.1995	7.4767	9.0859	10.8
6	4.631	4.58	4.7243	5.0533	5.5556	6.2435	7.163	8.3435	9.7556	11.3
7	4.9411	4.891	5.0402	5.3874	5.9309	6.6725	7.6184	8.7725	10.1309	11.6
8	5.1311	5.0811	5.2311	5.5808	6.1304	6.8797	7.8294	8.9797	10.3304	11.8
9	5.2011	5.1511	5.3011	5.6511	6.2011	6.9511	7.9011	9.0511	10.4011	11.9

Interpolated Isochronal Map

Use hover to inspect values. Zoom, pan, and reset are available in the toolbar.

Interpolated Isochronal Map



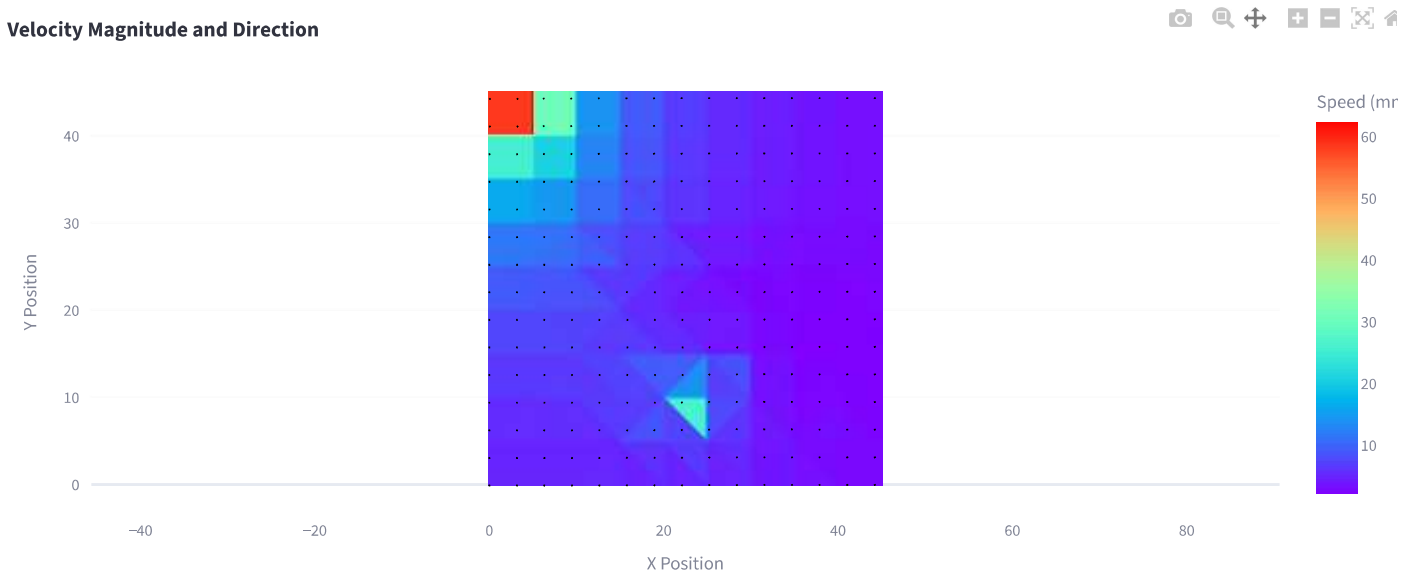


Download Isochronal Map (JPG)

Download Isochronal Map (SVG)

Velocity Field Map

The background shows speed magnitude; arrows show propagation direction.



Download Velocity Map (JPG)

Download Velocity Map (SVG)